CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (EE)

April/May 2012

EE 301 Electrical Machines - II

Date: 30.04.2012, Monday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 State the different methods of speed control for three phase induction motor and explain any one method in detail. [07]

Q - 2 (a) Explain the methods of improving the starting torque of three phase induction motor. [07]

(b) Explain different methods for measurement of slip of three phase induction motor. [07]

OR

Q - 2 (a) Write a note on shaded pole motor and universal motor. [07]

(b) Write a note on stepper motor and its application. [07]

Q - 3 (a) Explain cogging and crawling in poly phase induction motor. [07]

(b) Explain with suitable diagram how flux is rotating at synchronous speed. (production of 3-phase rotating magnetic field) [07]

OR

Q - 3 (a) The power input to a 6 pole, 50 Hz, 3 phase induction motor is 700 W at no load and 10 kW at full load. The no load copper losses may be assumed negligible, while the full load stator and rotor copper losses are 295 W and 310 W respectively. Find the full load speed, shaft torque and efficiency of the motor assuming rotational and core losses to be equal. [07]

(b) A 440 V, 3 phase, star connected induction motor has a stator impedance of (0.06 + j 0.2) ohm and an equivalent rotor impedance of (0.06 + j0.22) ohm. Determine the maximum gross power output and the slip at which it occurs. Exciting current may be neglected. [07]

SECTION – II

Q - 4 Explain the construction ,working, starting and running performance, characteristics and applications of capacitor-start capacitor-run single phase induction motor. [07]

OR

Q - 4 Why single phase induction motor is not self starting? Explain the operation of single phase induction motor by double revolving filed theory. [07]

Q - 5 Draw the circle diagram for a three phase, 6 pole, 50 Hz, 400V, star connected induction motor from the following datas (line values): No load test : 400V; 9 A, 1250 Watts. Short circuit test : 200 V, 50 A, 6930 Watts. The stator copper loss at standstill is 55 % of total copper losses and full load current is 32 A. From the circle diagram determine (i) power factor, slip, output, efficiency, speed and torque at full load (ii) starting torque (iii) maximum power factor (iv) maximum power output (v) maximum power input (vi) maximum torque in synchronous watts and slip for maximum torque. [14]

Q - 6 (a) Discuss the procedure for determining the parameters of equivalent circuit of single phase induction motors. Also draw an equivalent circuit for it. [07]

(b) Explain the deep bar & double cage poly phase induction motor with relevant diagrams. Also draw the equivalent circuit for double cage motor. [07]

OR

Q - 6 (a) Explain principle, working and application of induction generator. Draw the characteristics of induction generator showing the effect of capacitor. [07]

(b) Explain the function of commutator as a frequency changer. [07]

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